Tip: the journey bar is clickable - jump back to any finished stage, or between your tests, anytime.

RESULTS

Results

01 · CHI-SQUARE GOODNESS-OF-FIT

do counts match an expected split?

Table 1. Observed vs. expected

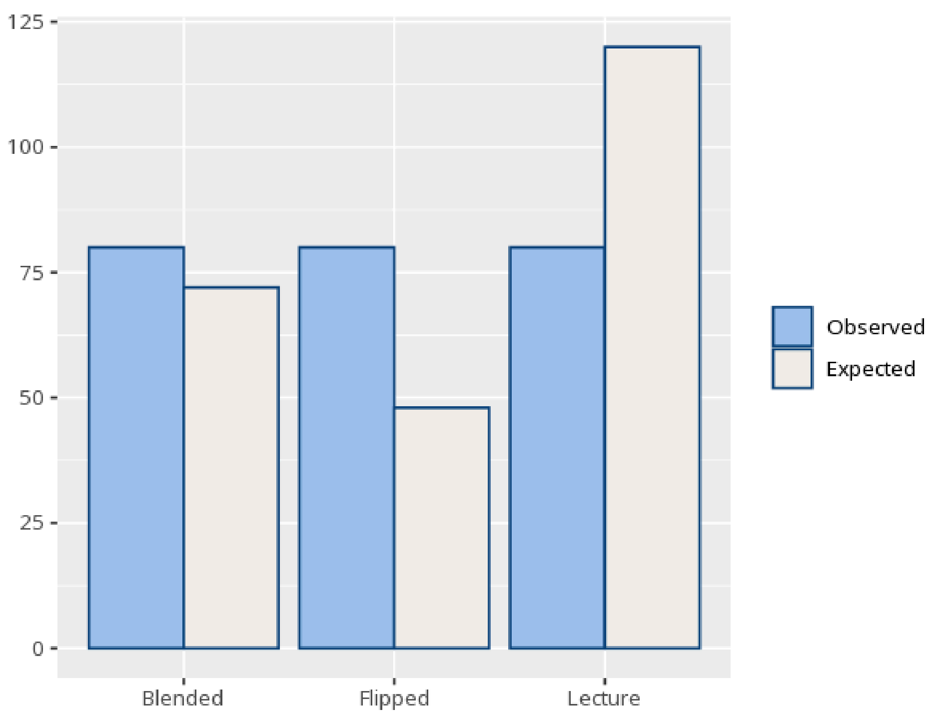
Category	Observed	Expected	Std. residual
Blended	80	72.00	1.13
Flipped	80	48.00	5.16
Lecture	80	120.00	−5.16

Table 2. Goodness-of-fit test

χ^2	df (k−1)	p	Cohen's w [95% CI]
35.56	2	<.001	0.38 [0.27, 2.00]

here df = number of categories − 1 (not (r−1)(c−1) as in the independence test).

Figure. Observed vs. expected



Understanding the numbers

Category. Each level of the variable being tested against its expected split. Here, 3 categories are compared to their expected counts.

Observed. The actual count seen in that category. Observed counts here range 80 across the 3 categories tested.

Expected. The count that category would have under the expected distribution (equal split, by default, unless custom proportions are set). Expected counts here range 48.00–120.00 across the 3 categories.

Std. residual. How many standard errors that category's observed count sits from its expected count; beyond about ±1.96 flags a category that significantly drives the result. Std. residual magnitudes here range 1.13–5.16 - values beyond about 1.96 flag the categories driving this result. The smallest expected count here is 48.0.

χ^2. The total discrepancy between observed and expected counts, summed across categories. Here, $\chi^2(2, N=240) = 35.56$.

df. Degrees of freedom here - the number of categories minus one (not $(r-1)(c-1)$ as in the independence test). Here, df = 2.

p. The probability of a split this different (or more different) from expected if the true proportions matched. Here, $p < .001$ - below your significance threshold, the observed split differs from expected.

Cohen's w. The size of the effect - how far the observed split departs from expected, independent of sample size. Here, w = .38 [.27, 2.00].

How to read this test

Tests whether one categorical variable's counts depart from an expected distribution (equal by default). A p below alpha means the observed split differs from expected; a standardized residual beyond about ±1.96 flags a category that significantly drives the result. Valid only when expected counts are mostly ≥ 5 (some texts allow ≥ 1 if no more than 20% are below 5); for sparse categories use an exact / simulated test instead. Your significance threshold (α) is 0.05.

APA template: A goodness-of-fit test, $\chi^2(2, N=240)=35.56$, $p < .001$, w=.38 [.27, 2.00].

Statistical basis: Chi-square goodness-of-fit test (Pearson, K. (1900). "On the criterion that a given system of deviations from the probable in the case of a correlated system of variables is such that it can be reasonably supposed to have arisen from random sampling." Philosophical Magazine, 50(302), 157-175.); Cohen's w effect-size benchmark (Cohen, J. (1988). Statistical Power Analysis for the Behavioral Sciences (2nd ed.). Routledge.). Full references in the exported CITATIONS.txt.

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